

Derek Wang

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EDUCATION

University of Waterloo | *Bachelor of Applied Science in Computer Engineering (Co-op)*

Expected April 2030

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Prandtl Dynamics Inc.

Toronto, ON

- Tracked 5+ simultaneous DJI drones at 170 m with 20 ms latency from a single microphone array via a Python pipeline chaining SRP-PHAT localization, per-source beamforming, and Unscented Kalman Filter tracking
- Raised acoustic drone detection from 85% to 98% mAP on a held-out split, with 40% fewer false alarms at half the size, by replacing the YAMNet audio classifier with EfficientAT in PyTorch
- Closed cross-tenant data exposure in 6 dashboard services via middleware org-scoping every query and WebSocket room
- Automated cleaning of 50+ hours of training audio with a Python tool that localizes and beamforms array recordings

Software Engineer Intern

March 2026 – June 2026

Kenoxis Technologies Inc.

Mississauga, ON

- Lowered API latency 35% by caching posting lookups and auth sessions in Redis across TypeScript/Node.js services
- Cut job-posting refresh time from daily to hourly by designing the MongoDB schemas and ingestion pipelines behind it
- Flagged suspicious applicants 60% faster by running LLM fraud screening and document review on async job queues

Autonomy Software Developer

September 2025 – Present

WATonomous – Student Design Team

Waterloo, ON

- Enabled 2 Hz obstacle-aware rover replanning via C++ ROS 2 A* on a 30x30 m LiDAR costmap and 10 Hz Pure Pursuit
- Led YOLO11s object detection retraining on 3,916 images from two merged datasets so the rover can spot a hammer, water bottle, and pickaxe at 0.96 mean average precision (mAP50)
- Accelerated YOLO11s inference to 30 FPS on the rover's NVIDIA Jetson Nano via ONNX export and onnxslim

PROJECTS

Mr. Clean – LLM-Orchestrated Mobile Manipulation | *Python, MuJoCo, PyTorch, Claude API* [GitHub](#) | [Demo](#)

- Drove a simulated two-arm robot across 3 tables from one typed prompt by building a Claude commander that hands each table to its own controller: an LLM skills agent, an imitation-learning policy, or a fly-connectome policy
- Sorted 4/4 cubes in 13 tool calls via a nested Claude agent whose 6 typed skills refuse unknown objects at the boundary
- Parked within 10 cm on 9/9 test routes with zero falls by following A* paths smoothed into clamped cubic splines
- Steered the base to 20/20 unseen goals with a policy wired from a fruit fly's brain (512-neuron FlyWire connectome)

7-DOF Robotic Arm Motion Control | *C++17, Eigen, MuJoCo, CMake, React, TypeScript*

[GitHub](#)

- Wrote 7-DOF arm kinematics and dynamics from scratch in C++ using Lie groups and recursive Newton-Euler, matching MuJoCo on 1,000 test poses for forward kinematics and inverse dynamics
- Held tip error under 0.2 mm on 46/49 targets with zero actuator saturation via jerk-limited 1 kHz computed torque
- Removed arm vibration and a 2 cm tolerance by moving inverse kinematics out of the 1 kHz loop to replan on demand
- Kept the elbow clear of obstacles and joint limits while holding the tool within 0.2 mm via null-space projection

Autonomous Vehicle State Estimation & Tracking | *C++, Eigen, ROS 2, Python, Docker, KITTI*

[GitHub](#)

- Localized a car to 0.558 m mean error over 6.6 km of KITTI driving, 2.3x tighter than its GPS input, via sensor fusion of 100 Hz IMU and 10 Hz GPS in a 15-state error-state Kalman filter in C++/Eigen
- Reached 0.886 multi-object tracking accuracy (MOTA) on ground-truth detections via an interacting multiple model filter blending 4 motion models, with Jonker-Volgenant matching on 3D IoU

Reparo – AI Repair Assistant | Hack Canada 2026 Winner (\$5,000) | *Python, FastAPI, Swift, Gemini*

[GitHub](#)

- Turned one device photo into a structured repair plan in under 60 s via component detection and Gemini reasoning

TECHNICAL SKILLS

Languages: C++, Python, MATLAB, Java, JavaScript, TypeScript, SQL, Swift

Frameworks/Libraries: ROS 2, PyTorch, OpenCV, Eigen, MuJoCo, ONNX, React, Next.js, Node.js, FastAPI

Infrastructure: Linux, Docker, GitHub Actions, CMake, Git, SQLite, MongoDB, Redis, Foxglove

Robotics: Motion Planning, State Estimation, Sensor Fusion, Computer Vision, Imitation Learning